Instructions:

- There are 3 questions, each 10 marks, for a total of 30 marks. Total time available 30 minutes.
- Provide answers only in the space provided. You may do your rough work on a separate blank sheet.
- You may use a calculator, but nothing else. **Pl. keep your phones/laptops, books etc. away from you**.
- Do not use unfair means, else action will be taken. Do not use material such as books, notes, etc. or a computer, smart phone, etc. And do not share information with other students.

$\boxed{10}$

1. Alice & Bob plan to use ElGamal Cryptosystem to exchange messages confidentially. Bob sends to Alice a message, M, suitably encrypted using ElGamal Cryptosystem that itself uses Diffie-Hellman generated shared key, K. The underlying Diffie-Hellman parameters are: prime **q = 19,** and its primitive root **a = 10.** While Alice and Bob select private keys, **XA** & **XB**, unknown to an intruder, Alice & Bob share with each other public keys **YA = 3** and **YB = 11.**
What is the message, M, given that the intruder has access to the computed cipher, C = 9? That is, what is M, given: q=19, root a=10, public keys YA=3, YB=11, cipher C=9. Show all details.

(all calculations are mod 19.)

Given $q = 19$, $a = 10$, $YA = 3$, $YB = 11$, $C = 9$ (cipher C)

3 (i) What is XA, given YA = 3. $YA = a^{XA} = 10^{XA} \Rightarrow XA = 5$

or

What is XB, given YB = 11. $YB = 11 = a^{XB} = 10^{XB} \Rightarrow XB = 6$

(ii) K (shared key) $= (YB)^{XA}$ or $(YA)^{XB}$

3 $\Rightarrow K = 11^5 = 7$, ~~or~~ (since $K = (YB)^{XA}$), OR

$K = (YA)^{XB} = 3^6 = 7$

(iii) Given cipher $C = 9 = K * M = 7 * M$.

4 compute K^{-1}. Since $K * K^{-1} = 1$ or $7 * K^{-1} = 1$

$\Rightarrow K^{-1} = 11$

Given cipher $C = 9$, $K^{-1} = 11$, $M = K^{-1}C = 11 * 9 = 4$

10

2. Device **A** wishes to send text, **T1**, to device **B**, and expects device **B** to reply with an ACK, **A1**. Device **B** will accept text, **T1**, from **A** and then reply to it with an ACK, **A1**, provided it is able to verify the origin & integrity of text sent by device **A**. Similarly, device **A** will accept the ACK from device **B** only if it is able to (i) verify the origin and integrity of the reply, **and** (ii) the ACK makes a reference to the earlier message or text, **T1**.
We assume devices **A** and **B** use an RSA-based public-key cryptosystem. They both have access to the Certification Authority, **CA**, that is responsible for creating public-key certificates for devices **A**, **B** or others.

Write below:

a. What is minimum information about Certification Authority, **CA**, that device, **A** or **B** should possess?

2 A, and similarly B, should possess the public key of CA, viz. KPU_{CA}

b. What specific information does device **A** seek from Certification Authority, **CA**? And for what purpose?

2 A seeks the public-key CERTIFICATE OF B, so as to obtain the public key of B, viz. $KPU\text{-}B$

c. What specific information does device **B** seek from Certification Authority, **CA**? And for what purpose?

2 B seeks public-key certificate of A, so as to obtain the public key of A, viz $KPU\text{-}A$.

d. What is the structure & content of the overall message that **A** sends to **B**, together with text **T1**, hash value (if any), encryption (if any) and Nonce (if any)? See example below.

2 $A \rightarrow B: E(K_{PR\text{-}A}, [T1 \| Nonce1])$

e. What is the structure & content of the overall message that **B** sends to **A**, together with ack, **A1**, hash value (if any), encryption (if any) and Nonce (if any)? See example below.

2 $B \rightarrow A: E(K_{PR\text{-}B}, [ACK1 \| f(Nonce1)])$

(As an example, a device X may send a message to Y using a message structure such as that given below:

$X \rightarrow Y: E(K_C, [K_B \| ID_{TGS}]) \| Ticket$

10

3. A portion of the **Kerberos v4** protocol is given below.

(1) $C \rightarrow AS$ $ID_c \| ID_{tgs} \| TS_1$

(2) $AS \rightarrow C$ $E(K_c, [K_{c,tgs} \| ID_{tgs} \| TS_2 \| Lifetime_2 \| Ticket_{tgs}])$

$Ticket_{tgs} = E(K_{tgs}, [K_{c,tgs} \| ID_C \| AD_C \| ID_{tgs} \| TS_2 \| Lifetime_2])$

(a) Authentication Service Exchange to obtain ticket-granting ticket

(3) $C \rightarrow TGS$ $ID_v \| Ticket_{tgs} \| Authenticator_c$

(4) $TGS \rightarrow C$ $E(K_{c,tgs}, [K_{c,v} \| ID_v \| TS_4 \| Ticket_v])$

$Ticket_{tgs} = E(K_{tgs}, [K_{c,tgs} \| ID_C \| AD_C \| ID_{tgs} \| TS_2 \| Lifetime_2])$

$Ticket_v = E(K_v, [K_{c,v} \| ID_C \| AD_C \| ID_v \| TS_4 \| Lifetime_4])$

$Authenticator_c = E(K_{c,tgs}, [ID_C \| AD_C \| TS_3])$

(b) Ticket-Granting Service Exchange to obtain service-granting ticket

(5) $C \rightarrow V$ $Ticket_v \| Authenticator_c$

(6) $V \rightarrow C$ $E(K_{c,v}, [TS_5 + 1])$ (for mutual authentication)

$Ticket_v = E(K_v, [K_{c,v} \| ID_C \| AD_C \| ID_v \| TS_4 \| Lifetime_4])$

$Authenticator_c = E(K_{c,v}, [ID_C \| AD_C \| TS_5])$

Pinpoint exactly which message, and specifically what content in the message, allows:

A. The Authentication server (AS) to authenticate the client, C?

2 Message 2
Message is encrypted by shared key K_c.

B. The client, C, to authenticate the authentication server AS?

2 Message 2
Message is encrypted by shared key K_c.

C. The Ticket-granting server TGS to authenticate Client, C?

2 Message 3
$Authenticator_c$, encrypted by key $K_{c,tgs}$

D. The Client, C, to authenticate the Ticket-granting server, TGS?

2 Message 4,
itself encrypted by shared key $K_{c,tgs}$

E. The Client, C, to authenticate the sever, V?

2 Message 6,
itself encrypted by shared key $K_{c,v}$